# Ali AGHABABAEI

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## Summary

I am a final-year Ph.D. candidate at Université Grenoble Alpes, supervised by Professor Massih-Reza Amini. My research explores model efficiency and representation learning within machine learning and deep learning, with the goal of designing scalable and practical AI systems.

## Education

**Université Grenoble Alpes** *2023 - Present*
*Ph.D. in Computer Science*

- **Research:** Deep Learning Efficiency, Representation Learning

**Sharif University of Technology** *2018 – 2022*
*M.Sc. in Computer Science*

- **Coursework:** Deep Learning, Speech Processing

**University of Tehran** *2013 – 2018*
*B.Sc. in Electrical Engineering*

- **Coursework:** Linear Algebra, Pattern Recognition, Digital Signal Processing

## Research Interests

**Deep Learning Efficiency**

- Tensor Decomposition
- Knowledge Distillation
- Memory-Efficient Training

**Representation Learning**

- Semi-Supervised Learning
- Pseudo-Labeling
- Contrastive Learning

**Generative AI**

- Probabilistic Modeling
- Image Denoising
- Image Super-Resolution

## Research Experience

**Ph.D. Thesis** *2023 - Present*
*APTIKAL team*

- Developing a new method for **memory-efficient** training by improving greedy supervised learning. Proposed a new method for preserving information during local training using neural information estimation.
- Extend the **Forward-Forward** algorithm by scaling it to Vision Transformers, introducing early cross-attention between label tokens and feature representations.
- Investigated semi-supervised learning, focusing on **pseudo-labeling** through a theoretical analysis of bias stability. Proposed an adaptive **Gaussian geometric** loss to shape latent representations, accelerate convergence, and reduce majority-class bias.
- Reduced the computational complexity and parameter count of pre-trained deep learning models by developing a **tensor decomposition** approach, utilizing a custom **constraint-based** loss function to identify optimal ranks while preserving accuracy.

**Research Collaborator** *2021 – 2022*
*University of Basel*

- Conducted analysis of inverse problems, including image denoising, limited-view computed tomography (CT), and wave scattering, utilizing invertible neural networks.
- Approximated data distribution with normalizing flows to identify out-of-distribution samples, improving model robustness and reliability.

**M.Sc. Thesis** *2018 – 2021*
*Electronic Research Institute*

- Proposed a patch-wise feature analysis approach for identifying forgery in video frames, aimed at enhancing deepfake detection capabilities.
- Developed a straightforward method to enhance the generalization and robustness of deepfake detection models.

**B.Sc. Thesis** *2016 – 2017*
*University of Tehran*

- Designed and implemented a video quality meter to assess video quality in terms of blockiness and blurriness distortions in a no-reference mode.

## Publications

**JEPAMatch: Geometric Representation Shaping for Semi-Supervised Learning** *Mar 2026*
***Ali Aghababaei-Harandi***, Aude Sportisse, Massih-Reza Amini
arXiv preprint

**Unified Framework for Pre-trained Neural Network Compression via Decomposition and Optimized Rank Selection** *Sep 2025*
***Ali Aghababaei-Harandi***, Massih-Reza Amini
ECML-PKDD 2025

**Deep Variational Inverse Scattering** *Mar 2023*
AmirEhsan Khorashadizadeh, ***Ali Aghababaei-Harandi***, Tin Vlašić, Hieu Nguyen, Ivan Dokmanić
EuCAP 2023

**Conditional Injective Flows for Bayesian Imaging** *Feb 2023*
AmirEhsan Khorashadizadeh, Konik Kothari, Leonardo Salsi, ***Ali Aghababaei-Harandi***, Maarten de Hoop, Ivan Dokmanić
IEEE Transactions on Computational Imaging

## Talks & Presentations

**Efficient Deep Learning** *2025*
Mathematical Fundamentals of AI Workshop, Sorbonne Center for AI (SCAI), Paris, France

**Presentation on Unified Framework for Pre-trained Neural Network Compression** *2025*
European Conference on Machine Learning (ECML-PKDD), Porto, Portugal
Conférence sur l'Apprentissage Automatique (CAp), Dijon, France

## Skills

**Programming:** Python, PyTorch, TensorFlow, JAX, Hugging Face, Git

**Theoretical:** Pattern Recognition, Design and Analysis of Algorithms, Deep Learning